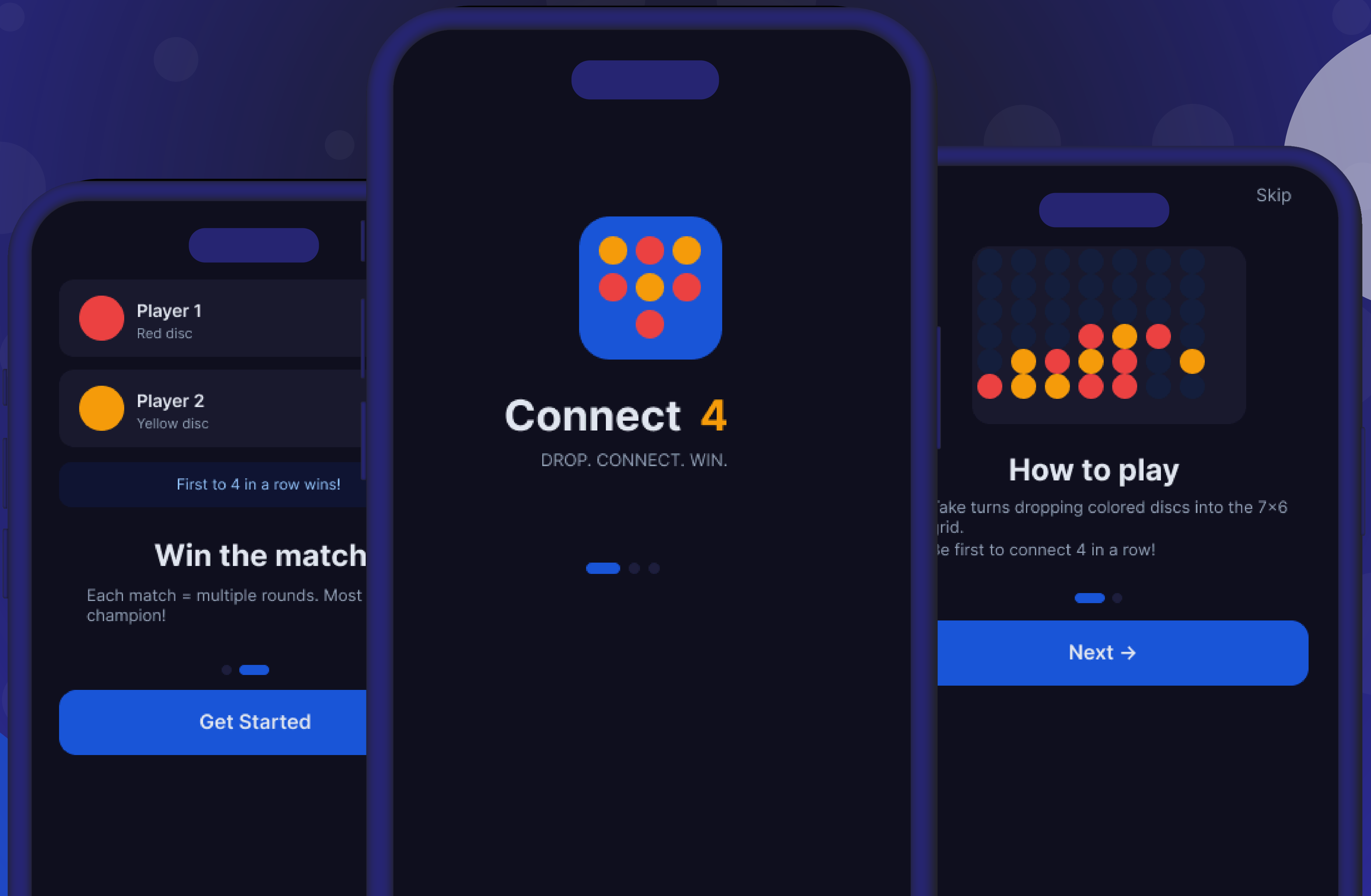
Connect Four

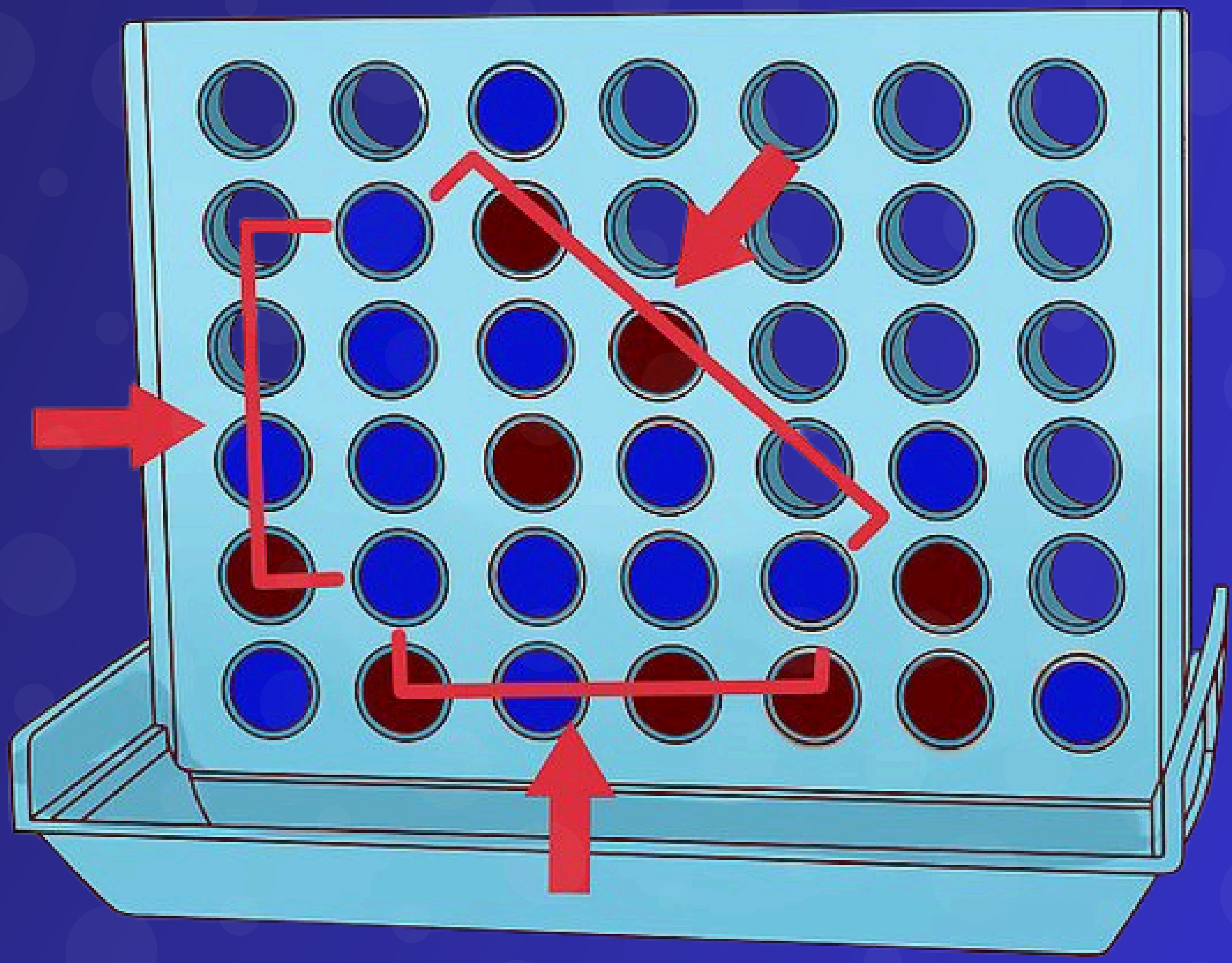


IDEA

- THE GAME IS ABOUT A 6 ROWS X 7 COLOUMNS MATRIX.
- THERE ARE TWO PLAYERS PLAYING AGAINST EACH OTHER -EACH WITH A UNIQUE COLOR-.
- EVERY PLAYER MUST MAKE ONE MOVE ONLY IN HIS TURN.
- IN ONE MOVE THE PLAYER CAN CHOOSE A COLOUMN WHERE HE CAN DROP HIS DISC.
- THE FIRST PLAYER TO MAKE ANY OF THE FOLLOWING WINS THE GAME:

  - MAKES 4 DISCS IN ROW VERTICALLY
  - MAKES 4 DISCS IN ROW HORIZONTALLY
  - MAKES 4 DISCS IN A ROW DIAGONALLY

- IF THE GRID IS FULL WITHOUT ANYONE WINNING THEN IT'S A TIE



PEAS

Performance

- **Optimization:** Does the agent make the optimal move or not?
- **Speed:** How much time does the agent spend thinking?
- **Mistakes:** How many mistakes does the agent make in one game?

Acuator

- In his turn he can chose a column to put into

Sensor

- A number representing the board state (bit board)

Environment

- The matrix
- The discs -what are the colors of the discs for both players-
- The current turn -who can make the move-

ODESA

Observability

Fully Observable

Episodic

Sequential

Determinisity

Strategic

Static

Semi-dynamic

Agent

Multi-agent
(Competitive)

TYPE

The agent is goal based

- We use the minimax algorithm to find out the best way to achieve goal